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EE1060: Differential Equations and Transform Techniques

Lecture 29: Convolution

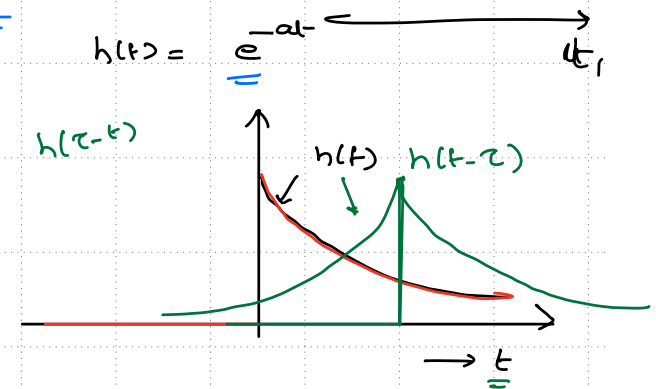
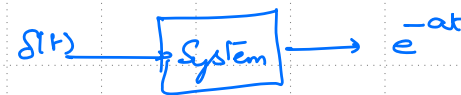
Topics :

1. Time Invariance
 2. Convolution
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Convolution for LTI Systems

Response of a LTI System to $x(t)$ is $x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$. $\leftarrow$
 $\downarrow$
 def for any two functions.

Example from L28:-

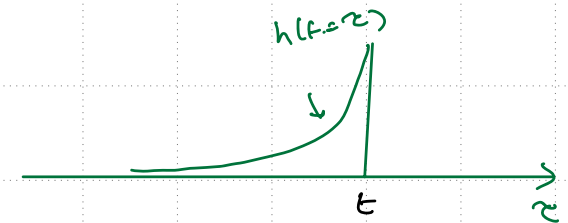


Causal system: resp is present when there is a driving source
 :- a system when the o/p depends on the past & current input
 & not a future inputs.

for a causal system, $h(t) = 0$ for $t < 0$.

Returning to example from L28 $\left\{ \begin{array}{l} h(t) = e^{-at} \\ h(t-\tau) = \end{array} \right.$

$$h(t-\tau) = \begin{cases} e^{-a(t-\tau)} & \tau < t \\ 0 & \tau > t \end{cases}$$



$$x(t) = \sin \omega t \, u(t) \\ x(\tau) \neq 0 \text{ for } \tau > 0$$

$$\Rightarrow y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau = \int_0^t \sin(\omega \tau) e^{-a(t-\tau)} d\tau$$

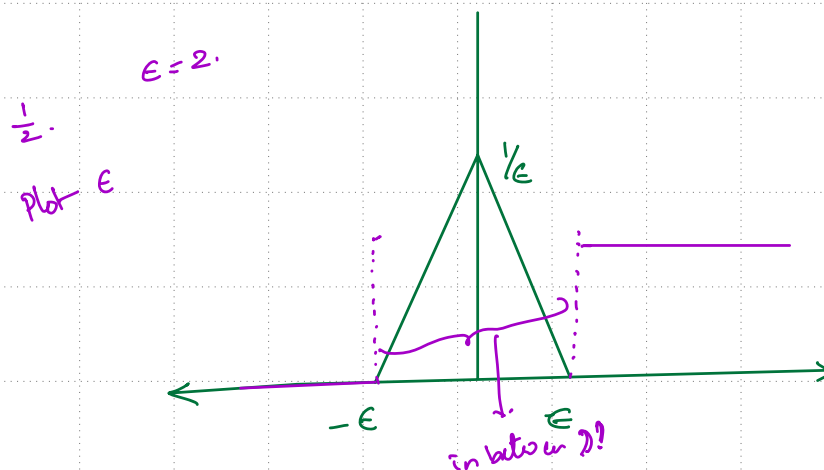
Example

Compute $y(t)$ when $x(t) = \sin \omega t \, u(t)$ and $h(t) = u(t)$
 $\rightarrow$ an integrator (can be used to get average / area under the curve)

$$h(t-\tau) = \begin{cases} 1 & \tau \leq t \\ 0 & \tau > 0 \end{cases}$$

$$y(t) = x(t) * h(t) = \int_0^t \sin \omega \tau \, d\tau = -\frac{\cos \omega \tau}{\omega} \Big|_0^t = \frac{1}{\omega} (1 - \cos \omega t)$$

Example 2:- $x(t) = \Delta^{1\text{cm}}$ and $h(t) = u(t)$; Compute $y(t)$? (graphically)



going by previous example, when $h(t) = u(t)$ then
 $y(t) = \int_0^t x(\tau) \, d\tau$ ← area under curve from 0 to t .

$$y(t) = \begin{cases} 0 & \text{for } t < -\epsilon \\ ?? & \text{for } |t| \leq \epsilon \\ \epsilon/2 & \text{for } t > \epsilon \end{cases}$$

H.W:- Compute the response analytically when $|t| < \epsilon$.

Example

Properties of Convolution :-

a) Commutative : $x(t) * h(t) = h(t) * x(t)$

b) Associative : $x * (h_1(t) * h_2(t)) = (x(t) * h_1(t)) * h_2(t)$

c) Distributive : $x(t) * (h_1(t) + h_2(t)) = x(t) * h_1(t) + x(t) * h_2(t)$

d) Scaling : $a \cdot x(t) * h_1(t) = a (x(t) * h_1(t))$

Response of an LTI system to e^{st}

Response of an LTI system when $x(t) = e^{st}$ where $s \in \mathbb{C}$

if $s = a + jb$ then, $x(t) = \underbrace{e^{at}}_{\substack{\text{growing} \\ \text{decaying}}} \underbrace{e^{jbt}}_{\text{oscillating}}$

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Let $h(t)$ be the impulse response of the system.

$$y(t) = x(t) * h(t) \quad (\text{or}) \quad h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau) e^{s(t-\tau)} d\tau$$

$$= e^{st} \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau$$

Scaling factor

O/p is scaled version of input

Laplace T/F
(to be discussed later).

$\rightarrow e^{st}$ is an eigen function of a LTI System.